DEWDRIVE · DESIGN PRECIS

DewDrive design precis

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DOCUMENT CONTROL

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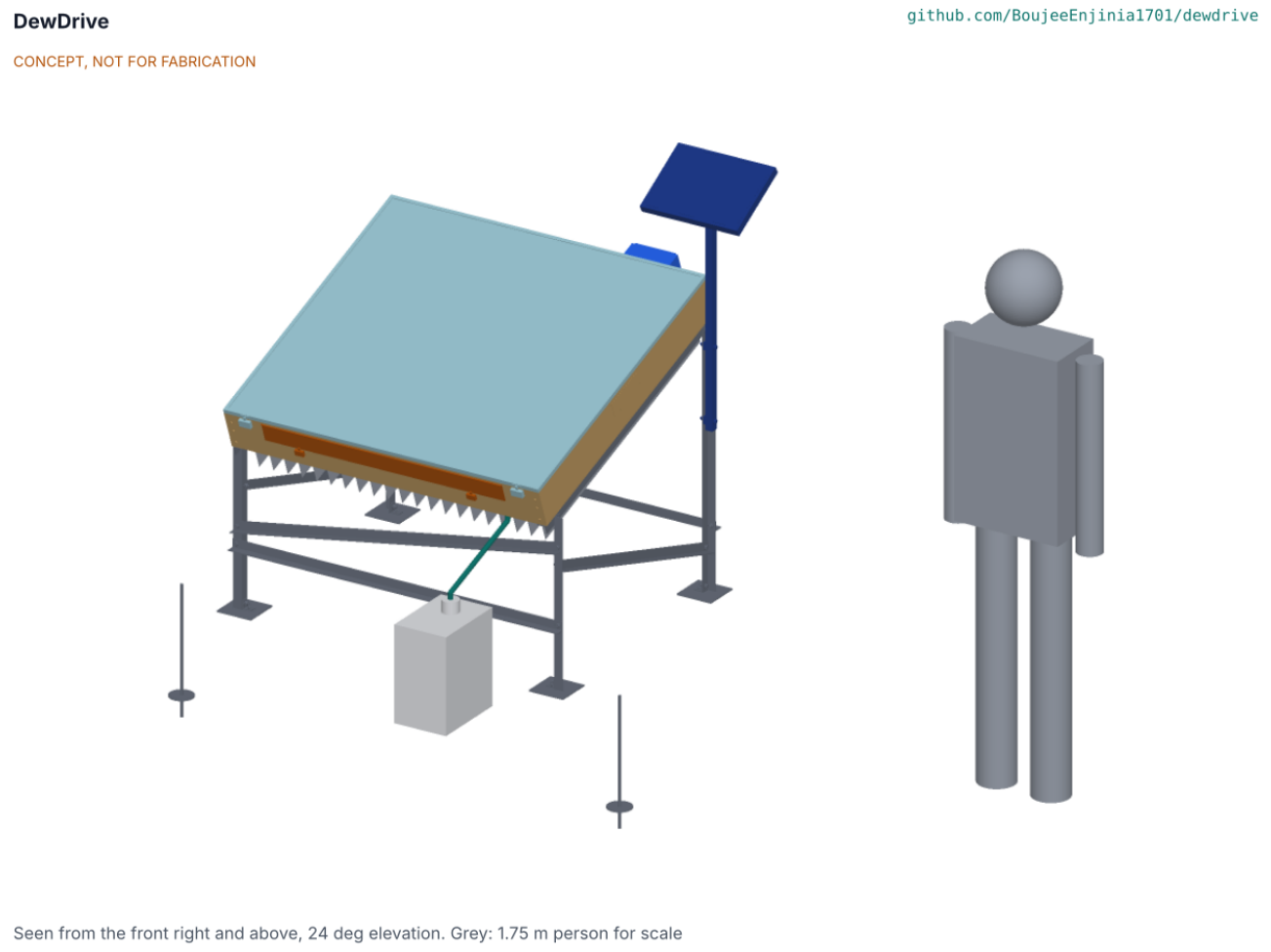
REVISION HISTORY

Version	Date	Author	Change
1.1	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: fan rule 65 % RH, restart below 60 % (item 3); design wind a 25 m/s gust (item 4)
1.0	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: STEM check fixes D16 to D20 (black absorber plates on the beds, PTFE-coated brine-wetted parts, open tray floor, painted condenser underside, 250 N anchors); PR-001 not met; numbers from DWD-CAL-001 v0.8; links to the STEM check, concept options and product requirements
0.9	2026-10-02	Amish Chadha	Follow-ups carried out: vent study concluded (low-emissivity screens dropped), label and 130 °C glazing in the BOM, cost USD 639 against the USD 520 target
0.8	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried in (DWD-DEC-001 items 2 to 5): glazing rated 130 °C or more, stagnation vent study, R9 restated with lifting rule, first partner type

0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-10-01	Amish Chadha	Design made constructable (DWD-DDR-003): tray deck, one outlet through the fan hood, bolted stand; mass and cost from DWD-CAL-001 v0.4
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); through-flow trays, 25 wt % salt, drip screens, fan humidity cut-out, anchors as default; numbers from DWD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 update (design choices decided by Amish in DWD-DDR-001; numbers replaced by DWD-CAL-001; parametric model and drawing DWD-DWG-001; yield shortfall and proposed fixes)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (how it works, components, first-order numbers, design choices, safety, open questions, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive design precis

DewDrive is a glazed, insulated box 1.1 x 1.0 m, tilted 20° toward the sun on a steel stand. Inside, four black trays hold 4 kg of silica gel impregnated with 25 wt % calcium chloride, each bed covered by a black perforated absorber plate. At night the vent flaps are opened and a 2 W fan draws air in above the trays and down through their mesh floors, and the sorbent takes up water. In the morning the flaps are shut; the sun heats the absorber plates and the bed to about 105 °C by noon and drives the vapour down past a black drip screen onto a finned aluminium floor that forms the condenser, shaded by the box itself. The condensate runs down the slope into a gutter and a 10 L bottle. The TRL 3 calculation (DWD-CAL-001 v0.8) gives about 0.44 L per day at 40 % night RH and 0.31 L at 25 %: the 0.25 L dry-night target is met, but the 0.5 L design-point target is not, and how to recover it is awaiting Amish (DWD-DEC-001, open item 2). The design follows Amish's decisions of 2026-09-25 (DWD-DDR-002), which replaced the v0.1 layout that made only 0.20 L per day. Made buildable in DWD-DDR-003 and corrected by the first STEM check (DWD-STM-001; DWD-DDR-005), the estimated parts cost is $695 against the $520 value-engineering target (moved to $520 by Amish on 2026-09-26), $175 over the target. The STEM check, the three concept options and the product requirements are in DWD-STM-001, DWD-OPT-001 and DWD-PRS-001; the concept choice between options A and B, which score within 10 %, is awaiting Amish (DWD-DDR-004). One unit supplements drinking water; it does not replace a water supply.



Seen from the front right and above, 24 deg elevation. Grey: 1.75 m person for scale

Figure 1. DewDrive on its stand, with a 1.75 m person for scale. Glazing faces south (for a northern-hemisphere site), the collection bottle sits below the low edge and the 10 W PV panel is on a pole to the north, where it cannot shade the box. Generated from the parametric model `cad/src/model.py`.

How it works

1. **Adsorb at night.** In the evening the user opens the south inlet flap, which sits above the tray deck, and the outlet flap over the fan on the north wall. The logger switches on the fan, which draws 40 m³/h of night air in over the trays, down through the 9.3 mm sorbent beds and their mesh floors, and out under the deck through a slot into the fan hood, for 10 h. The trays sit on an EPDM gasket on the sealing baffle of a lift-out deck, so the air cannot bypass the beds. The $CaCl_2$ in the silica gel pores binds water first as hydrates and then as a solution held in the pores; the silica gel adds a little uptake of its own and keeps the solution in place. If the air goes above 65 % RH the logger stops the fan, and restarts it below 60 %.
2. **Seal and heat by day.** In the morning the user shuts both flaps. Sunlight passes through the twin-wall polycarbonate and heats the black perforated absorber plates that rest on the beds (the beads themselves are white and would reflect most of it). The bed passes 83 °C by 09:00 and reaches about 105 °C at noon, and its vapour pressure rises above that of the condenser.
3. **Condense.** Vapour diffuses down through the open tray floor and the 73 mm gap under the beds, around the drip-screen channels, to the aluminium floor plate, which stays within about 12 K of ambient thanks to 21 painted fins in the shade under the box.
4. **Collect.** Water films run down the 20° slope to a gutter along the low edge and drain through a silicone tube into a food-grade jerrycan.
5. **Log.** The logger records air temperature and RH, bed and condenser temperatures every 5 min, and applies the fan humidity cut-out; the user weighs the bottle each day. The data make yields comparable between sites.

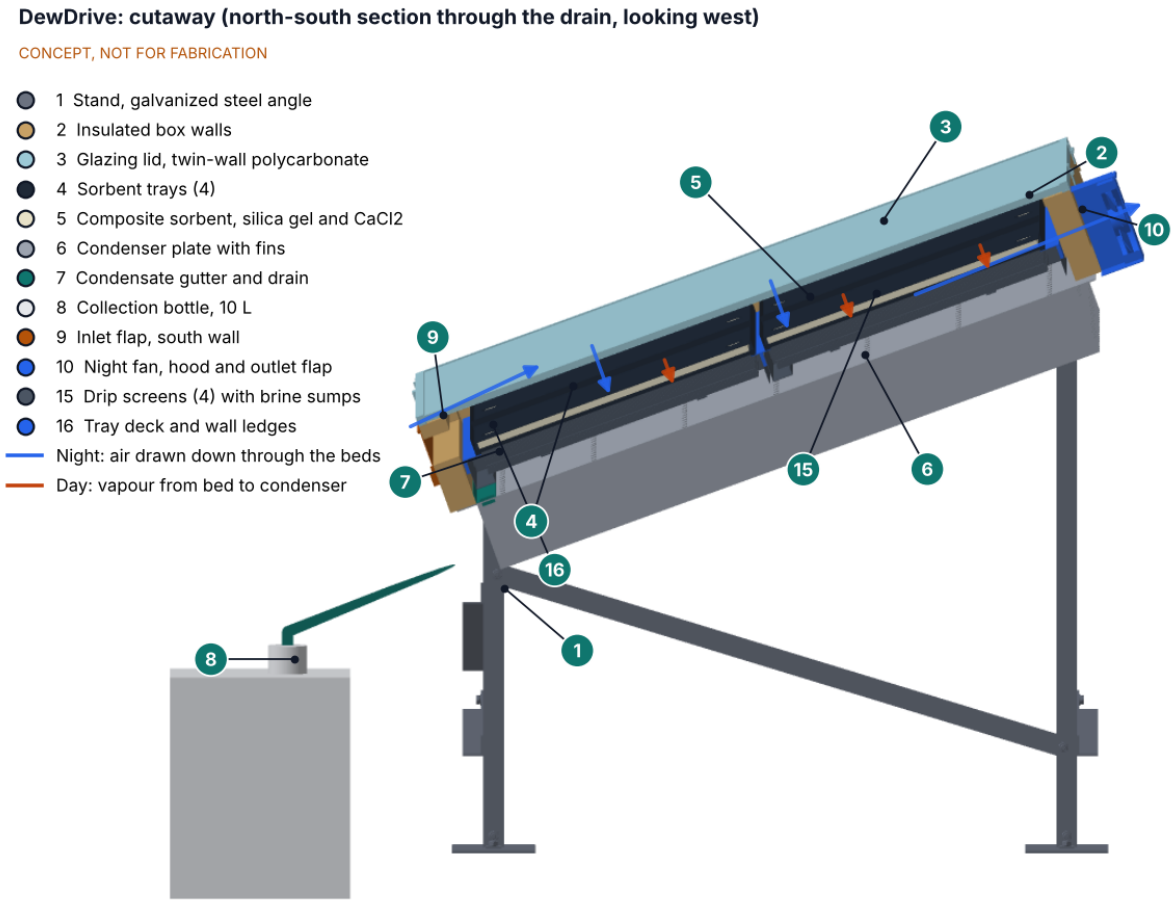


Figure 2. North-south section through the drain, looking west. Blue arrows: night air path with the flaps open, in above the trays and down through the beds. Red arrows: daytime vapour path from the bed (5) past the drip screen (15) to the condenser floor (6). The gutter (7) along the low edge drains to the bottle (8).

Main components

Table 1. Main components. Numbers match `bom/bom.csv`, drawing DWD-DWG-001 and Figures 2 and 3.

#	Component	Choice	Notes
1	Stand	Galvanized steel angle 30 x 30 x 3 mm, rails, legs, braces and cross members, bolted; tilt 20°	Separable from the box; two ground anchors rated 250 N or more each by default (D20), about 40 kg of ballast as the alternative (DWD-DDR-002, D13; DWD-CAL-001, H2 to H4)
2	Insulated box walls	Plywood skins with 25 mm PIR foam, 1,100 x 1,000 x 140 mm, 40 mm thick	Open top (glazing) and bottom (condenser)
3	Glazing lid	10 mm UV-stabilized twin-wall polycarbonate, hinged on the north edge	Rated 130 °C or more; a 120 °C sheet only if supplier data and the inner skin measured at the TRL 4 stagnation test both show it stays inside its rating (DWD-DEC-001, item 3)
4	Sorbent trays (4) with absorber plates	Aluminium pans 490 x 430 x 25 mm with floors of PTFE-coated glass-fibre mesh on seven 6 x 1.5 mm bars (D19); black PTFE inside, bare underside; a black PTFE-coated perforated aluminium absorber plate (3 mm holes at 6 mm pitch, 22.7 % open) on each bed; EPDM edge gasket on a 1 mm aluminium sealing baffle	Lift out for recharging; the baffle forces the night air through the beds (D9); absorbers, coatings and open floor from the STEM check (DWD-DEC-001 D16, D17, D19)
5	Composite sorbent	4 kg: 3.0 kg mesoporous silica gel with 1.0 kg $CaCl_2$ (25 wt %, D10)	Bed 9.3 mm deep on 0.785 m²; all food-grade
15	Drip screens (4)	Two staggered layers of 20 x 6 mm aluminium channels at 30 mm pitch under each tray, running down the slope to a closed brine sump; black PTFE-coated (D17)	Catch any brine that leaves the beds (sumps 0.58 L in all) with no line of sight from bed to condenser (D11)
6	Condenser	2 mm aluminium floor plate with 21 fins, 1 x 100 x 920 mm at 50 mm pitch	Wetted face food-safe coated; fins and underside painted with a high-emissivity paint (D18)
7	Gutter and drain tube	Gutter along the low edge; food-grade silicone tube	
8	Collection bottle	10 L HDPE jerrycan	About 17 days of production at the design point
9	Vent flap, south inlet	Hinged flap 800 x 40 mm above the trays, with EPDM seal and latches	Closed by day; seal quality sets the vapour loss
10	Night fan, hood and outlet flap	12 V 120 mm fan, about 2 W, in a hood over a slot in the north wall below the deck; the outlet flap closes the fan opening by day	Fan switched by the logger; stops above 65 % RH, restarts below 60 %
11	PV panel	10 W, 12 V, on a pole north of the box	Runs fan and logger
12	Electronics box	PWM charge controller, 12.8 V 6 Ah LiFePO4 with BMS, ESP32 logger	Only low-voltage DC on the device
13	Sensors	Air T and RH in a radiation shield; bed and condenser probes	The air RH reading drives the fan cut-out

Item 14 (hardware and consumables) is in the BOM; since kit 1.9 every screw, bolt, nut, washer, insert and rivet is modelled as its own part. Item 16 is the lift-out tray deck with its sealing baffle and the wall ledges. The general arrangement is drawing DWD-DWG-001 Rev P6 (`cad/drawings/`); the STEP files are in `cad/step/` (one per part in `cad/step/A/`); how to build it is the build plan DWD-BLD-001 (`docs/05-build-plan.md`).

Fan rule (DWD-DDR-002, D11; thresholds set by Amish on 2026-10-25, DWD-DEC-001 item 3). The logger runs the fan from the evening flap opening for 10 h, and stops it whenever the air RH is above 65 % (restarting below 60 %, to avoid cycling), 6 points below the 71 % RH at which the pores fill, so a low-cost sensor's tolerance cannot carry the bed past it. This is a design rule for the firmware sketch; firmware beyond a sketch is TRL 4 work and on hold.

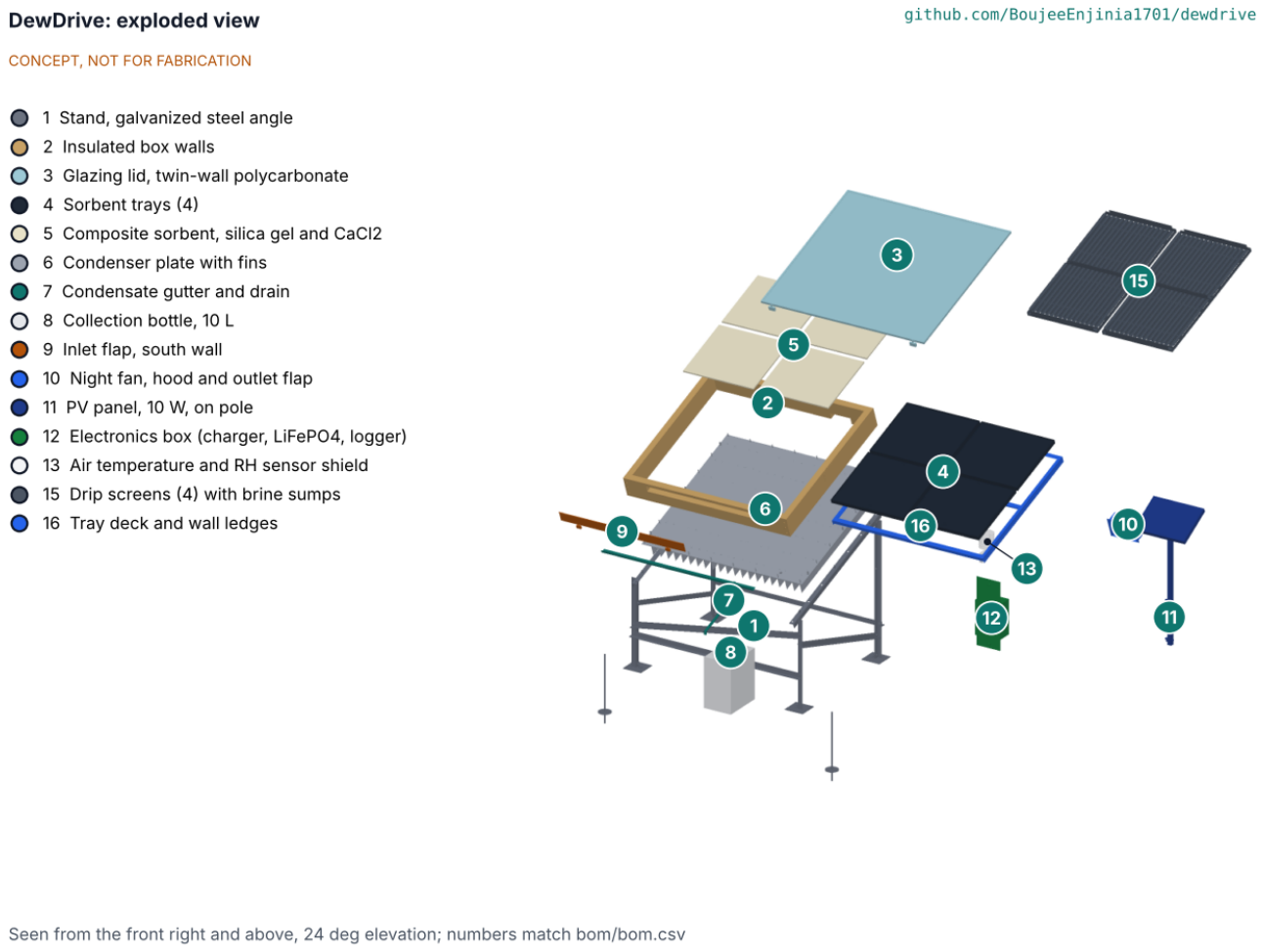


Figure 3. Exploded view with BOM numbers.

Numbers from DWD-CAL-001

All values are first-principles estimates from `docs/04-calcs/sizing.py` (DWD-CAL-001 v0.8); the tag after each value is the line of the script output that carries it. The sorbent isotherm is built from bulk salt data and must be replaced by measured isotherms of the actual composite.

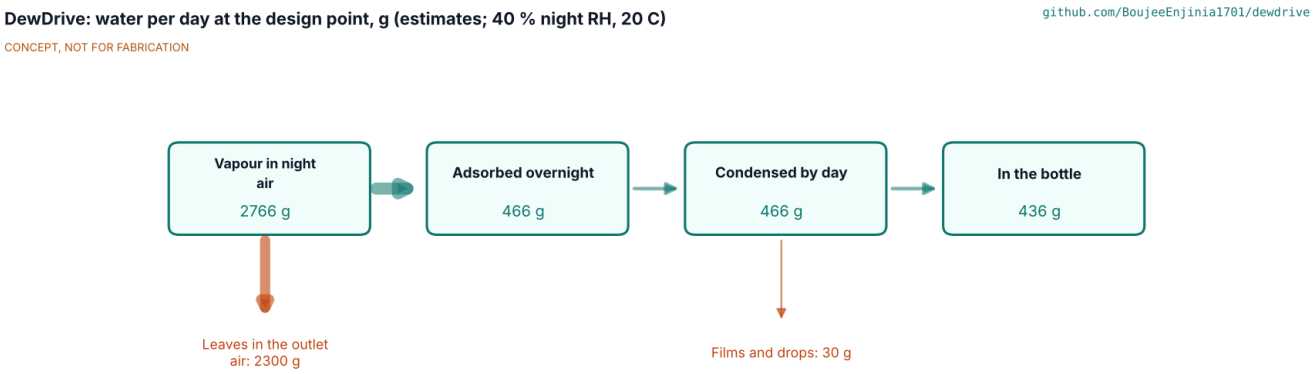


Figure 4. Water per day at the design point (40 % night RH, 20 °C). All values are estimates.

Table 2. Water, energy, size and cost.

Quantity	Value	Basis	Requirement
Vapour carried through the box overnight	2.77 kg (B4)	40 m³/h for 10 h at 6.92 g/m³	
Share of the vapour driving force captured	Nearly all (gas-side NTU 19, B3)	Air drawn down through the bed; 0.16 Pa pressure drop	v0.1 layout: 11 % (B2)
Overnight uptake	0.47 kg (B5)	Cyclic simulation; equilibrium would allow 1.59 kg (A4)	
Collected at 40 % night RH	0.44 L per day (C3)	0.47 kg condensed less 30 g of films; 0.37 to 0.41 L under the sensitivities (C9, C12, C13)	PR-001 not met
Collected at 25 % night RH	0.31 L per day (C4)	Same model	PR-002 met
Bed temperature at noon	105 °C (C2)	785 W/m² peak; transmittance-absorptance 0.70 with the absorber plates (C11)	PR-003 met
Condenser rise above ambient, peak	11.9 K (C2)	Load 292 W peak; UA 24.6 W/K with the painted underside (C8)	PR-005 met on the correlation; at risk
Where the sun goes	3.46 kWh absorbed; 0.33 kWh desorbs water; 1.53 kWh lost through the glazing; 1.52 kWh to the condenser (C7)	The black drip screens cut the effective bed-to-condenser emissivity from 0.802 to 0.405 (C5)	
Solar-to-water efficiency	5.4 % (C7)	Published devices reach about 9 % (LaPotin et al., 2021)	
Stagnation temperature, dry bed	111 °C (D1); 133 °C on a peak day, glazing inner skin 106 °C (D6)	Noon, 35 °C air; peak day 1000 W/m² and 40 °C	Glazing rated 130 °C or more (DWD-DEC-001, item 3)
Electrical use and supply	23.6 Wh per day against 35 Wh (F1); 2.6 days of autonomy (F2)	Fan 2 W for 10 h; logger 0.15 W	PR-004 met
Salt containment	Pores fill above 71 % RH (E2); 47 % full after one 90 % RH night (E3)	Sumps 0.58 L (E6)	PR-007 met on paper
Mass	61.1 kg (135 lb) dry; box 36.1 kg, 23.4 kg with the trays and deck lifted out (G1, G2)	Model components	PR-009 met as restated on 2026-10-02 (lift with the trays and deck out)
Wind, 25 m/s design gust (DWD-DEC-001 item 4)	495 N normal to the lid; tips without two anchors of 234 N each (specified 250 N) or 96 kg of ballast (H1 to H4); 317 N and 39 kg at the earlier 20 m/s (H5)	Force coefficient 1.2 on 1.1 m²	PR-010 not verifiable at TRL 3
Parts cost	$695 (J1)	`bom/bom.csv`	

Quantity	Value	Basis	Requirement
			PR-011 over the value-engineering target by $175 (target $520)
Water per dollar over 5 years	about $0.87 per litre (J2)	Parts only	For comparison

What the numbers say:

- **Drawing the air through the bed fixes the yield.** The v0.1 layout, with air skimming past the trays, captured only 11 % of the vapour driving force and made 0.20 L per day. Through the bed, the air leaves close to equilibrium with the sorbent, and the day desorption now sets the limit. If slow diffusion inside the grains cut the night NTU to 1, the yield would fall to 0.38 L per day (C9).
- **The absorber plates make the day work.** The white beads absorb only about 0.3 of the sun; without the black plates the design would make 0.01 L per day (C11). This was found by the first STEM check and fixed on 2026-10-25 (DWD-DEC-001, D16).
- **The tray floor must let the vapour out.** The independent review found that a perforated sheet under a fine mesh passed only about 25 % of the bed area, not the 60 % the calculation had taken; the floor is now a coarser mesh on thin bars, 59 % open (D19), which lifts the yield from 0.36 to 0.44 L per day. It still misses the 0.5 L target.
- **The drip screens help the yield.** Black channels under the trays catch brine and also cut the radiation from the hot beads to the condenser by about 40 %: 0.44 L per day with them against 0.31 L without (C6).
- **A low-emissivity screen would do more, but overheats the box.** Bare aluminium channels would give 0.82 L per day, but a dry bed would stagnate at about 159 °C, beyond the rating of polycarbonate glazing (C10, D1). That option is not adopted. Amish decided on 2026-10-02 (DWD-DEC-001, item 4) on one short paper study of a passive stagnation vent. The study (DWD-CAL-001, section D) finds that a vent of about 117 cm² would hold the dry bed at 122 °C, but it would have to stay shut in normal operation, when the bed already runs at 130 to 149 °C, so it needs a thermostat and is not passive or modest in cost. The low-emissivity screens are therefore dropped and the prototype keeps the black screens.
- **The lower salt loading contains the brine.** At 25 wt % the solution fills the pores only above 71 % RH, against 51 % RH at the v0.1 loading (E2).
- **Cost is over the value-engineering target once the design is buildable.** The gasket, baffle and drip screens added $27 (total $515, D15); the parts that make the design buildable (DWD-DDR-003) bring the estimated cost to $620; the 2026-10-02 decisions add a glazing sheet rated 130 °C or more and a label ($639); and the STEM check fixes of 2026-10-25 add the absorber plates, PTFE coatings, floor bars and underside paint, making $695 against the $520 target ($175 over).
- **Yield is still modest.** About 0.44 L per day is about one sixth of one person's survival drinking need.

Key design choices

Choices 1 to 7 were decided by Amish on 2026-09-25 (DWD-DDR-001, going with the recommendation), and choices 8 to 11 on the same day (DWD-DDR-002). Choice 12 was part of the design from the start.

1. **Single-stage, flat glazed box** rather than a dual-stage or tubular design (D4). Dual-stage could add about 20 % in later versions (LaPotin et al., 2021).

2. **Silica gel and $CaCl_2$ composite**, rather than MOF, zeolite or LiCl (D3), at 25 wt % salt (D10, amending the 33 wt % of D3).
3. **Condenser as the shaded floor of the box**, directly below the bed (D4). Short vapour path, no pump, one fewer seal.
4. **Fan-assisted night airflow** with a 10 W PV panel and a 12.8 V LiFePO4 battery, rather than natural airflow (D5).
5. **Manual flaps** opened and closed by the user, rather than automatic actuators (D6).
6. **20° tilt, fixed**, facing the equator, with twin-wall polycarbonate glazing rather than glass (D7).
7. **Design point** of 20 °C and 40 % RH at night and 6.0 kWh/m² of sun, with a 25 % RH low-humidity case (D8).
8. **Night air drawn down through the sealed trays** (D9), with the inlet flap above the trays and the fan below them.
9. **Drip screens and a fan humidity cut-out** for wet spells (D11). The screens are black, not low-emissivity, after the paper evaluation of D12.
10. **Two ground anchors by default**, ballast as the alternative (D13).
11. **Value-engineering target kept at $500** (D14), then moved to $520 by Amish on 2026-09-26 (D15) once the BOM reached $515.
12. **Built-in logging** so that each prototype yields comparable data.

Safety

Safety: Harvested water must be tested and treated before drinking. Condensed water is close to distilled, but it can pick up bacteria, dust, salt from the sorbent and metal from the condenser. Until a laboratory test shows otherwise, boil or disinfect it and do not use it as the only drinking source. Distilled-like water also lacks minerals.

Safety: The inside of the box gets hot enough to burn: an estimated 105 °C in the bed at noon and about 111 °C with a dry bed, 133 °C on a peak day (DWD-CAL-001 v0.8, C2, D1 and D6). The glazing outer skin reaches about 55 °C and the condenser fins about 47 °C. Open the lid only when cool, wear gloves when handling trays and drip screens, and label the glazing. The polycarbonate must be rated 130 °C or more (DWD-DEC-001, item 3); a 120 °C sheet is acceptable only if the supplier's data and the inner-skin temperature measured at the TRL 4 stagnation test both show it stays inside its rating. Do not fit bare, low-emissivity screens: they would raise the dry bed to about 159 °C.

Safety: Calcium chloride irritates eyes and skin and gives off heat when it dissolves. Wear gloves and eye protection when preparing the sorbent and when emptying the drip-screen sumps. At 25 wt % the solution fills the pores only above about 71 % RH (DWD-CAL-001, E2); the fan stops above 70 % RH and the drip screens catch any brine. Keep trays level, close the flaps in rain, and never let the solution reach the water path.

Safety: The 12.8 V LiFePO4 battery has a built-in BMS and must be fused at the battery. Keep the electronics box shaded, and do not charge a damaged or swollen battery.

Safety: The tilted lid catches the wind like a sail (495 N in the 25 m/s design gust). Without restraint the unit tips over (DWD-CAL-001, H2). Fit the two ground anchors, each rated 250 N or more in the site soil, or about 96 kg of ballast, before the box goes on the stand. Empty the box of its trays and deck before lifting it; it carries a label saying so (R9 as restated, DWD-DEC-001, item 2). Deburr all aluminium fins, channels and sheet edges; fins under the box are at ankle and hand height.

Open questions

- [ ] What are the measured uptake isotherms and rates of the 25 wt % composite at 20 °C and at 80 to 120 °C? These set every yield number.
- [x] How can the parts cost come back to $500, or should the value-engineering target move? Decided by Amish, 2026-09-26: target moved to $520 (DWD-DDR-002, D15).
- [ ] Can a passive stagnation vent hold a dry bed near 122 °C at a modest cost? Decided by Amish, 2026-10-02 (DWD-DEC-001, item 4): one short paper study; if it cannot, the low-emissivity screens are dropped.
- [x] Glazing rating with a 122 °C dry bed. Decided by Amish, 2026-10-02 (DWD-DEC-001, item 3): rated 130 °C or more, or a 120 °C sheet only with supplier data and a measured inner-skin temperature at the TRL 4 stagnation test.
- [ ] How much salt carries over into the condensate, and does the aluminium condenser need a coating or a stainless replacement?
- [ ] How well do simple flap and tray seals hold, by night against bypass air and by day against vapour loss?
- [ ] Which site provides the humidity and solar data for the design point? Partner type decided by Amish, 2026-10-02 (DWD-DEC-001, item 5): the first candidate to approach, not yet agreed, is a university dryland field station in a hot semi-arid region with night humidity of about 25 to 40 % and an existing weather record, with a water and sanitation NGO for the later community trial.